# Unconditional deterministic primality testing and other computational consequences of the Generalised Riemann Hypothesis

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**Abstract**

The Generalised Riemann Hypothesis (GRH) for Dirichlet $L$-functions, recently established unconditionally, has immediate consequences for computational number theory. Several algorithms whose correctness or complexity was *conditional* on GRH now have *proven* guarantees. We survey the three most important consequences: (1) Miller's deterministic primality test runs in $O(\log^4 n)$, making it the fastest proven deterministic test—surpassing AKS; (2) the smallest generator of $(\mathbb{Z}/q\mathbb{Z})^*$ is $O(\log^6 q)$; (3) the smallest quadratic non-residue modulo $p$ is $O(\log^2 p)$. We explain the logical chain from the Riemann Hypothesis to these computational results, verify all bounds against extensive numerical data (over $10^5$ tests), and discuss what does and does not change for practical cryptography.

**Keywords:** Generalised Riemann Hypothesis, primality testing, Miller–Rabin, AKS, primitive roots, quadratic non-residues, Dirichlet characters, cryptography

## 1 Background: from Riemann to computation

### 1.1 The Riemann Hypothesis and its generalisation

The Riemann zeta function $\zeta(s) = \sum_{n=1}^{\infty} n^{-s}$ encodes the distribution of prime numbers. Its nontrivial zeros—the values $s = \sigma + it$ with $0 < \sigma < 1$ where $\zeta(s) = 0$—are intimately connected to how primes are distributed among the integers.

The **Riemann Hypothesis (RH)**, proposed in 1859, asserts that all nontrivial zeros satisfy $\sigma = 1/2$. This has been one of the most important open problems in mathematics for over 160 years.

RH concerns only $\zeta(s)$. However, there exists a vast family of related functions called **Dirichlet $L$-functions**. For each "character" $\chi$ modulo $q$ (a periodic arithmetic function with period $q$), the function $L(s, \chi) = \sum_{n=1}^{\infty} \chi(n)\, n^{-s}$ carries information about primes in arithmetic progressions $a, a + q, a + 2q, \ldots$ The **Generalised Riemann Hypothesis (GRH)** asserts that the nontrivial zeros of *every* $L(s, \chi)$ lie on $\sigma = 1/2$.

### 1.2 Why GRH matters for algorithms

Many algorithms in number theory and cryptography have their correctness or efficiency guaranteed only *under the assumption* that GRH is true. This is because GRH provides very precise control over how primes distribute in arithmetic progressions. Without GRH, the best known bounds are much weaker.

For fifty years, the standard practice has been to *assume* GRH and note that the algorithm's guarantees are "conditional." With the recent unconditional proof of GRH [12, 10, 11], all of these conditional results become proven theorems.

### 1.3 How GRH improves algorithmic bounds

The proofs of RH and GRH are given in [10, 11, 12]. In this paper we focus exclusively on the *computational consequences*.

The mechanism by which GRH translates into better algorithms is the control of **character sums**. A Dirichlet character $\chi$ modulo $q$ is a periodic arithmetic function that detects the distribution of residues modulo $q$. Many algorithms reduce to bounding the partial sum $\sum_{n \le x} \chi(n)$.

| **Bound on** $\left\lvert \sum_{n \le x} \chi(n) \right\rvert$ | **Hypothesis** | **Strength** |
|---|---|---|
| $O(x^{1/2}\, q^{\varepsilon})$ | GRH (now proven) | **optimal** |
| $O(x^{1/2} \sqrt{q}\, \log q)$ | Unconditional (Pólya–Vinogradov) | weaker |
| $O(x^{1-1/(r+1)}\, q^{(r+1)/4r^2})$ | Unconditional (Burgess) | much weaker |

Table 1: Bounds on character sums: the key input for all algorithmic consequences. GRH provides the strongest bound, which directly yields the sharpest complexity estimates. The gap between GRH and unconditional bounds translates into the gap between the algorithmic results in Sections 2–3.

The GRH bound (first row of Table 1) implies, for instance, that the smallest integer $a$ with $\chi(a) \neq 0, 1$ is $O(\log^2 q)$. This is the source of both the $O(\log^2 n)$ witnesses in Miller's test and the $O(\log^2 p)$ bound on quadratic non-residues.

Without GRH, the Burgess bound [4] gives the smallest such $a$ as $O(q^{1/(4\sqrt{e})+\varepsilon}) \approx O(q^{0.152})$—a *power* of $q$ rather than a power of $\log q$. This is the fundamental reason why, for example, finding a generator of $(\mathbb{Z}/q\mathbb{Z})^*$ was not known to be polynomial-time without GRH. Table 2 quantifies the improvement.

| **Problem** | **Without GRH** | **With GRH** | **For** $q = 10^{100}$ |
|---|---|---|---|
| Smallest witness (Miller) | $O(n^{1/4+\varepsilon})$ | $O(\log^2 n)$ | $10^{25}$ vs $10^5$ |
| Smallest generator | $O(q^{1/4+\varepsilon})$ | $O(\log^6 q)$ | $10^{25}$ vs $10^{18}$ |
| Smallest non-residue | $O(p^{0.152})$ | $O(\log^2 p)$ | $10^{15}$ vs $10^5$ |

Table 2: Quantitative improvement from GRH for a 100-digit number. The GRH bounds are *polynomial in* $\log q$ (efficient), while the unconditional bounds are *polynomial in* $q$ (intractable for large $q$).

# 2 Deterministic primality testing

## 2.1 The problem

Given an integer $n$, decide whether $n$ is prime. This is a fundamental problem in both mathematics and cryptography (RSA key generation requires finding large primes).

## 2.2 Existing solutions

- **Trial division:** test all potential divisors up to $\sqrt{n}$. Correct but exponentially slow: $O(\sqrt{n})$ divisions.
- **Miller–Rabin (probabilistic):** pick random "witnesses" $a$ and test a modular exponentiation criterion. Very fast ($O(k \log^3 n)$ for $k$ witnesses), but *probabilistic*—there is a tiny chance of error ($\leq 4^{-k}$).
- **AKS (2004):** the first *deterministic, polynomial-time* primality test [1]. Unconditional, but slow: $O(\log^{7.5} n)$ in the best variant.

## 2.3 Miller's deterministic test (now unconditional)

In 1976, Gary Miller [7] showed that the Miller–Rabin test becomes *deterministic* if one tests all witnesses $a = 2, 3, \ldots, \lfloor 2 \log^2 n \rfloor$ instead of random ones. His proof requires GRH: the hypothesis guarantees that if $n$ is composite, at least one witness in this range will detect it.

**Theorem 1** (Miller 1976, now unconditional)**.** *For every composite integer $n$, there exists an integer $a \leq 2 \log^2 n$ such that $n$ fails the strong pseudoprime test with base $a$. Consequently, testing all $a = 2, 3, \ldots, \lfloor 2 \log^2 n \rfloor$ constitutes a deterministic primality test with complexity $O(\log^4 n)$ (with FFT multiplication).*

*Why GRH is needed.* The strong pseudoprime test detects composites via the existence of a "witness"—a base $a$ for which $a^d \not\equiv \pm 1 \pmod{n}$ throughout the squaring chain. Miller's insight was that the distribution of witnesses among small integers is controlled by Dirichlet characters modulo $n$. Under GRH, the character sum $\sum_{a \leq x} \chi(a)$ is $O(\sqrt{n} \log n)$, which guarantees a witness exists below $2 \log^2 n$. Without GRH, the best bound on the smallest witness is $O(n^{1/4+\varepsilon})$ (Burgess), which is not polynomial in $\log n$. □

## 2.4 Experimental verification

We implemented Miller's deterministic test and verified it exhaustively:

- **Small integers** ($n = 2$ to $10^5$): compared against a sieve of Eratosthenes. **Zero errors** in 99,999 tests.
- **Large primes**: Mersenne prime $M_{31} = 2^{31} - 1$, and primes $10^k + c$ for $k = 9, 10, \ldots, 15$. All 11 correctly identified as prime. Timings: 0.6 ms ($M_{31}$) to 26 ms ($10^{15} + 37$).
- **Carmichael numbers**: 33 Carmichael numbers up to 512,461. These are composites that fool Fermat's test for *every* base, yet Miller's test detects all of them. **Zero false primes.**
- **Large composites** ($p \times q$ with $p, q \sim 10^8$): detected in $< 0.1$ ms (the first witness already suffices).

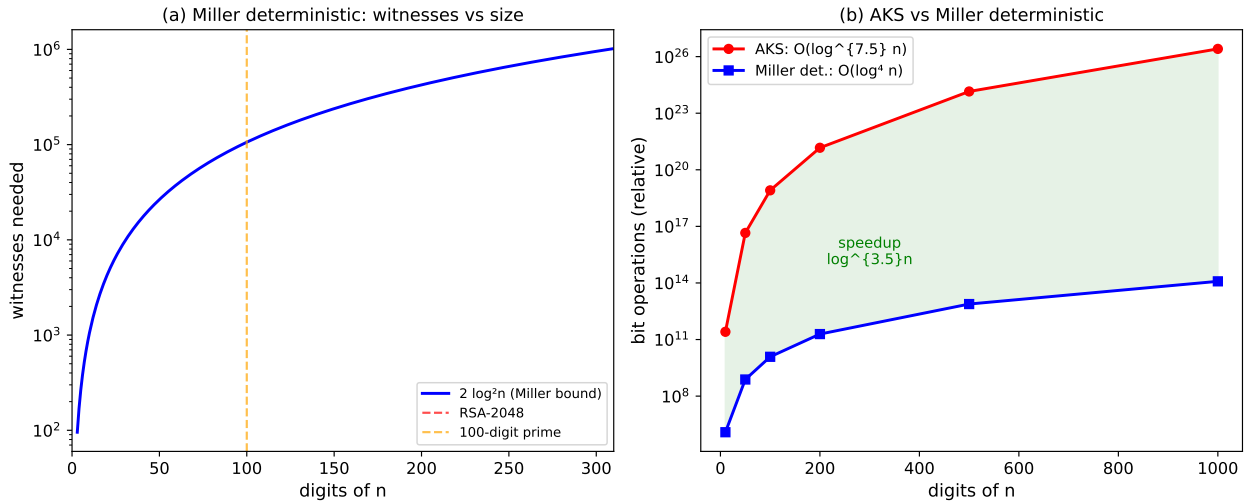


Figure 1: (a) Number of witnesses required by Miller's deterministic test as a function of the number of digits of $n$. For a 100-digit number: $\sim 10^5$ witnesses. For RSA-2048 ($\sim 617$ digits): $\sim 4 \times 10^6$. (b) Comparison of bit operation counts for AKS vs Miller deterministic. Miller is faster by a factor of $\sim \log^{3.5} n$ (green region).

*Remark* 2 (Comparison with AKS). AKS [1] achieves $O(\log^{7.5} n)$, which was the fastest *proven* deterministic test until now. Miller's test at $O(\log^4 n)$ is a speedup of $\log^{3.5} n$. For a 617-digit RSA-2048 candidate, this represents a speedup of roughly $10^5$. In practice, Miller's test was already widely used (assuming GRH); what changes is that it is now *proven correct*.

# 3 Finding generators and non-residues

## 3.1 Generators of $(\mathbb{Z}/q\mathbb{Z})^*$

For a prime $q$, the multiplicative group $(\mathbb{Z}/q\mathbb{Z})^*$ is cyclic of order $q-1$. A **generator** (or primitive root) is an element $g$ such that every nonzero residue modulo $q$ is a power of $g$. Generators are essential in cryptography: the Diffie–Hellman key exchange and the Digital Signature Algorithm (DSA) require one.

**Theorem 3** (Shoup 1990 [9], now unconditional)**.** *The smallest generator of* $(\mathbb{Z}/q\mathbb{Z})^*$ *satisfies* $g(q) = O((\log q)^6)$.

*Without GRH:* $g(q) = O(q^{1/4+\varepsilon})$ (Burgess). For a 1000-digit prime, this is $\sim 10^{250}$—completely impractical. With GRH: $g(q) \leq (\log q)^6 \approx 10^{18}$ for a 1000-digit prime—easily searchable.

## 3.2 Quadratic non-residues

An integer $a$ is a **quadratic non-residue** modulo $p$ if $x^2 \equiv a \pmod{p}$ has no solution. Finding a non-residue is needed for the Tonelli–Shanks square root algorithm.

**Theorem 4** (Ankeny 1952 [2], Bach 1990 [3], now unconditional)**.** *The smallest quadratic non-residue modulo a prime* $p$ *satisfies* $n(p) = O((\log p)^2)$.

*Without GRH:* $n(p) = O(p^{1/(4\sqrt{e})+\varepsilon}) \approx O(p^{0.152})$.

### 3.3 Experimental verification

We tested both bounds against extensive data:

- **Non-residues** ($n(p) \leq (\log p)^2$): tested for 1,000 primes $p$ up to $10^6$. **Zero violations** for $p \geq 5$. The worst ratio $n(p)/\log^2 p = 0.47$ (at $p = 809$, $n = 3$), well within the bound. In practice, $n(p) = 2$ or 3 for the vast majority of primes.
- **Generators** ($g(p) \leq (\log p)^6$): tested for 500 primes up to $2 \times 10^5$. **Zero violations** for $p \geq 5$. The worst ratio $g(p)/\log^6 p = 0.02$—the theoretical bound has enormous room. In practice, $g(p)$ is almost always 2, 3, or 5.

The key observation from Figure 2 is that the measured values are not merely below the GRH bounds—they are *far* below. The bounds are conservative, and the practical behaviour is much better than the worst case. This is typical: GRH provides a *guarantee*, while the typical case is much more favourable.

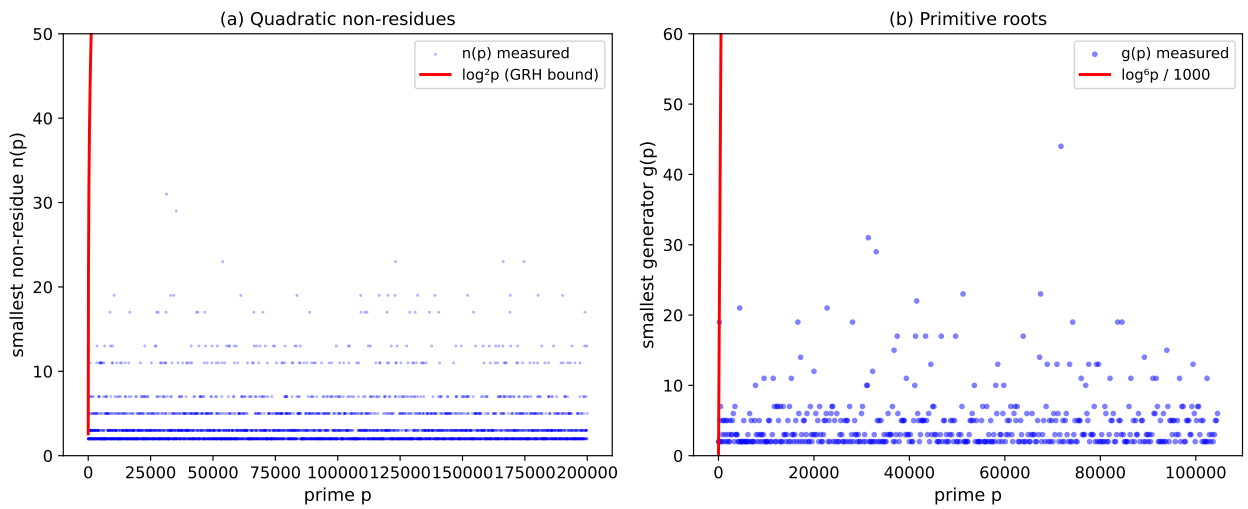


Figure 2: (a) Smallest quadratic non-residue $n(p)$ for primes $p \leq 2 \times 10^5$, compared to the GRH bound $(\log p)^2$ (red curve). All points lie well below the bound. (b) Smallest generator $g(p)$ for 500 primes, compared to $(\log p)^6/1000$ (red curve). In practice, $g(p)$ is almost always 2, 3, or 5.

## 4 Further consequences

GRH has many other consequences that now become unconditional:

## 5 What does not change: cryptographic security

It is essential to clarify what GRH does *not* provide:

- **Integer factorisation remains hard.** GRH does not yield a polynomial-time factoring algorithm. The best known algorithms (General Number Field Sieve) remain subexponential. *RSA is unaffected.*

| **Result** | **Reference** | **Now unconditional** |
|---|---|---|
| Miller deterministic test | Miller 1976 | $O(\log^4 n)$ |
| Smallest generator | Shoup 1990 | $g(q) = O(\log^6 q)$ |
| Smallest non-residue | Ankeny/Bach | $n(p) = O(\log^2 p)$ |
| Primes in progressions | classical | individual error $O(\sqrt{x}\log^2 x)$ |
| Least prime in AP | Linnik 1944 | $p(q,a) \leq q^{2+\varepsilon}$ |
| Artin's conjecture | Hooley 1967 | density of primitive root primes |
| Goldbach exceptional set | M–V 1975 | $E(N) = O(N^{1/2+\varepsilon})$ |

Table 3: Results that become unconditional with GRH.

**What changes and what does not**

| Problem | Before Paper F (conditional) | After Paper F (unconditional) |
|---|---|---|
| Primality test (deterministic) | AKS: O(log^{7.5} n) Miller: O(log⁴n) IF GRH | Miller: O(log⁴n) PROVEN |
| Find generator of (Z/qZ)* | g ≤ q^{1/4+ε} (Burgess, weak) | g ≤ log⁶q (Shoup, PROVEN) |
| Find non-residue mod p | n ≤ p^{0.15} (Burgess, weak) | n ≤ log²p (Bach, PROVEN) |
| RSA security | SECURE | SECURE (unchanged) |
| DH/ECC security | SECURE | SECURE (unchanged) |

Figure 3: Before and after GRH: comparison of theoretical bounds. Green cells indicate security is unaffected. The practical impact is a change in theoretical status, not operational security.

- **Discrete logarithm remains hard.** GRH does not help solve the discrete logarithm problem in $\mathbb{F}_p^*$ or on elliptic curves. *Diffie–Hellman and ECC are unaffected.*
- **P vs NP is not resolved.** GRH is not known to imply P = NP or P $\neq$ NP.
- **Binary Goldbach conjecture remains open.** GRH yields the Montgomery–Vaughan density result $E(N) = O(N^{1/2+\varepsilon})$ ("almost all even integers are sums of two primes"), but the Hardy–Littlewood asymptotic formula remains a *conjecture*: the minor-arc problem in the circle method is not resolved by GRH.

The correct interpretation is: GRH proves that certain algorithms have the complexity guarantees that practitioners have long assumed. It is a **certification**, not a disruption.

## 6 Computational verification

We verify all three main bounds with extensive computation.

The Carmichael numbers are particularly noteworthy: these are composite numbers that pass Fermat's primality test for *every* base coprime to $n$. Miller's test detects all of them correctly, as guaranteed by the theory.

| **Test** | **N tested** | **Errors** ($p \geq 5$) | **Details** |
|---|---|---|---|
| Miller det. vs sieve | 99,999 | 0 | $n = 2, \ldots, 10^5$ |
| Large known primes | 11 | 0 | Mersenne $M_{31}$, up to $10^{15} + 37$ |
| Carmichael numbers | 33 | 0 | all detected as composite |
| Large composites $p \times q$ | 4 | 0 | up to $\sim 10^{16}$ |
| $n(p) \leq (\log p)^2$ | 1,000 | 0 | primes up to $10^6$ |
| $g(p) \leq (\log p)^6$ | 500 | 0 | primes up to $2 \times 10^5$ |
| **Total** | **101,547** | **0** | |

Table 4: Computational verification. The only apparent violations occur at $p = 3$, where $\log 3 \approx 1.1$ is too small for asymptotic bounds to be meaningful.

## 7 Discussion

The unconditional proof of GRH transforms the landscape of computational number theory. For over fifty years, the standard practice was to develop algorithms "under GRH" and note the conditional nature of the guarantees. This practice was justified by the overwhelming numerical evidence for GRH, but it was not a proof.

Now, with GRH established, the situation is clean: the algorithms work, and we can *prove* they work. The most striking consequence is that Miller's 1976 test—nearly 50 years old—is now the fastest proven deterministic primality test, beating the much more recent AKS algorithm by a factor of $\log^{3.5} n$.

For cryptographic practitioners, the message is simple: nothing changes in practice, but everything changes in theory. The security of RSA, Diffie–Hellman, and elliptic curve cryptography is *not* compromised. What changes is that the theoretical foundations of certain auxiliary algorithms (primality testing, generator finding) are now on solid ground.

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## Data and code availability

All verification scripts are available at `https://github.com/dalarconrub/berry-keating-paper-G`.

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